Does LayerNorm Cause Creative Collapse? Decoder-Dependent Effects of Mean-Subtraction on Divergent Association in Language Models

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Abstract

Generative language models reliably collapse onto common, "human-intuitive" outputs in creative tasks. A widely cited mechanistic suspect is LAYERNORM, which forces hidden states onto a fixed-radius sphere and removes magnitude as a representational degree of freedom. We test this hypothesis directly: does relaxing LAYERNORM at inference change concept-association geometry and the Divergent Association Task (DAT) score? Across three pretrained Pre-LN Transformers (GPT-2, GPT-2-MEDIUM, PYTHIA-410M), three decoding strategies (greedy, top-p, high-temperature), seven LAYERNORM intervention recipes, and four layer-range targets, we measure (i) per-layer concept-pair cosine distance, (ii) DAT scores, and (iii) perplexity as a fluency check. We find that only RMSNORM-equivalent ablation ("NO_MEAN") preserves fluency; all stronger ablations break the model. Removing mean-subtraction shifts final-layer concept-pair distance by $+7\%$ on PYTHIA-410M, confirming the predicted geometric expansion. The behavioural effect on DAT, however, is small and decoder-dependent: under greedy and moderate-temperature top-p decoding, LAYERNORM relaxation does not move DAT in any consistent direction. Only at temperature $T{=}1.2$ does the geometric expansion propagate, raising DAT by $+1.7$ to $+5.1$ points across all four layer-range targets in GPT-2 (Cohen's d up to $+0.69$). We further document an artifactual "DAT inflation" failure mode in which broken interventions emit out-of-vocabulary tokens that score spuriously high. We conclude that LAYERNORM is one of several gates between latent variation and output diversity, but not the dominant one — the LM-head and softmax recompress most of the freed latent variation.

1 Introduction

Modern language models routinely generate fluent text but reliably collapse onto common, "human-intuitive" outputs in creative tasks. Ask a base GPT-2 for ten unrelated nouns and the same shortlist appears across seeds; ask an instruction-tuned model and the homogenisation is even sharper. This collapse is the observable face of *concept incongruence*: many valid combinations exist in latent space, but the generation process contracts onto the mode.

Two mechanisms are usually implicated. The first is *decoding*: top-p and temperature sampling deliberately suppress the distribution's tails. The second is *post-training*: instruction tuning and RLHF further sharpen the conditional distribution toward preferred answers. A third candidate — much less explored empirically — sits inside the architecture itself: LAYERNORM. Recent geometric analyses establish that LAYERNORM projects hidden states onto a hyperplane orthogonal to the uniform vector $\mathbf{1}$ and rescales them onto a fixed-radius sphere [Brody et al., 2023, Gupta et al., 2024, Menary et al., 2024]. Magnitude is removed as a representational degree of freedom,

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and concept subspaces are coupled through a shared denominator. If LAYERNORM itself contributes to creative collapse, this is a *mechanistic* lever independent of decoder or post-training intervention.

Gap. The literature has done parts of the puzzle but has not bolted them together. Geometric work [Brody et al., 2023, Gupta et al., 2024, Menary et al., 2024] establishes what LAYERNORM does to representations. Behavioural work [Olson et al., 2021, Chen and Ding, 2023, Bellemare-Pepin et al., 2024, Nagarajan et al., 2025] establishes that the Divergent Association Task (DAT) cleanly measures associative creativity. *No prior work measures* DAT *under controlled* LAYERNORM *ablations.* This paper fills that gap.

Approach. We hook every `nn.LayerNorm` in three pretrained Pre-LN Transformers (GPT-2, GPT-2-MEDIUM, PYTHIA-410M), replace its forward pass with one of seven recipes (STOCK, NO_MEAN, NO_AFFINE, WEAKENED_A with $\alpha \in \{0.25, 0.5, 0.75\}$, and IDENTITY), apply each to four layer ranges (ALL, EARLY, MID, LATE), and measure: (i) per-layer concept-pair cosine distance among 20 unrelated nouns; (ii) DAT scored against GLOVE-6B-300d on iterative one-noun-per-step generation; (iii) perplexity on a calibration text. We sweep three decoding strategies (greedy, top-p with T=0.7, high-temperature with T=1.2). The final-block LAYERNORM is preserved in all conditions, following Kanavalau et al. [2026]'s "scale anchor" result.

Findings preview. Only NO_MEAN (i.e., RMSNORM-equivalent) preserves fluency; all stronger ablations inflate perplexity by $10^2 - 10^5\times$ and emit gibberish. Within the fluency-preserving cells, removing mean-subtraction increases final-layer concept-pair distance by $+7\%$ on PYTHIA-410M (Wilcoxon paired test, Δ=$+0.026$). The behavioural DAT shift, however, is *decoder-dependent*: under greedy and moderate-T top-p, no cell moves DAT in a consistent direction. Only at T=1.2 does the geometric expansion propagate, with all four NO_MEAN targets pushing DAT by $+1.7$ to $+5.1$ points and Cohen's d up to $+0.69$. Under Bonferroni correction across 18 comparisons, no individual cell reaches significance, but the unanimous direction of the high-T panel is itself the signal. We also document an artifactual "DAT inflation" failure mode: when interventions break the model, the surviving outputs are out-of-vocabulary tokens with random embeddings that score spuriously high on cosine distance.

Contributions.

- We provide the first direct measurement of DAT under controlled LAYERNORM ablations, across three Pre-LN models, three decoders, seven intervention recipes, and four layer ranges (2,600 noun generations per (model, decoder) pair).
- We show that the geometric expansion predicted by Gupta et al. [2024] *does* appear at the final layer of PYTHIA-410M ($+7\%$ pair distance) but is decoder-gated at the behavioural level: DAT shifts only emerge under high-temperature sampling.
- We identify a DAT-specific failure mode where broken models score spuriously high via out-of-vocabulary tokens, motivating validity-rate gating as a co-equal sanity check.
- We localise the dominant collapse mechanism away from LAYERNORM: even when LAYERNORM is relaxed enough to expand latent geometry, the LM-head and softmax recompress most of the freed variation. LAYERNORM is one of several gates between latent variation and output diversity, but not the dominant one.

The paper proceeds as follows. section 2 reviews LAYERNORM geometry and DAT-based creativity measurement. section 3 details the intervention framework, prompts, and statistics. section 4 reports perplexity, DAT, and per-layer geometry. section 5 interprets the cross-modality reconciliation, lists limitations, and relates findings to prior work. section 6 concludes.

2 Related Work

Geometric role of LAYERNORM. Brody et al. [2023] decompose $\text{LayerNorm}(\boldsymbol{x}) = \gamma \odot (\boldsymbol{x} - \mu)/\sigma + \beta$ into a projection (onto the hyperplane orthogonal to $\mathbf{1}$) and a scaling onto a fixed-radius sphere, and show that both components are needed for attention: the projection enables uniform-attention queries (3× faster majority computation), and the scaling fixes the *unselectable-key* problem of Demeter et al. [2020]. Gupta et al. [2024] extend this to seven LLMs (including PYTHIA-410M, GPT-J, and LLAMA) and show that hidden states already have near-zero mean before LAYERNORM, so the mean subtraction is empirically near-redundant — which makes RMSNORM [Zhang and Sennrich, 2019] a natural drop-in. Menary et al. [2024] formalise the cross-coupling

of semantic subspaces caused by Pre-Norm's shared denominator, and report that O(10%) L2-norm perturbations collapse $\sim 1\%$ of attention heads. Singhal and Kim [2025] find divergent roles in Pre-LN versus Post-LN: in Pre-LN, removing LAYERNORM both destabilises learning and increases memorisation, with early-layer LAYERNORM most influential. None of these studies measures the consequence for creative-divergence metrics.

Norm-removal architectures. A growing literature replaces or removes per-token normalisation: NGPT [Loshchilov et al., 2025] forces every vector to unit norm; TAPERNORM [Kanavalau et al., 2026] introduces a gated cosine-decay from RMSNORM to a fixed linear map and identifies the *final* normalisation as the critical scale anchor; DYT [Zhu et al., 2025] replaces LAYERNORM with dynamic element-wise nonlinearities. These works study perplexity and downstream accuracy, not divergent association. Our intervention recipes (NO_MEAN, NO_AFFINE, WEAKENED_A, IDENTITY) are inference-time analogues of these training-time designs.

Creativity in LLMs. Olson et al. [2021] introduce DAT as an objective measure of associative creativity in humans, scoring the mean pairwise cosine distance among 7 of 10 unrelated nouns via GLOVE. Chen and Ding [2023] adapt DAT to LLMs, finding that GPT-4 (greedy) scores 89.1, beating 96% of $8{,}572$ human respondents. Bellemare-Pepin et al. [2024] confirm with $100{,}000$ humans across DAT and creative writing. Nagarajan et al. [2025] introduce algorithmic-creativity tasks (Sibling/Triangle/Circle/Line) and report that *seed-conditioning* (input randomness) beats temperature for diversity — pointing to internal-state, not output-state, mechanisms. Our work asks whether LAYERNORM is one such internal-state mechanism. He et al. [2025] probe creativity layer-wise via Layer-Skip and find an early-layer optimum for the hallucination-creativity tradeoff. Our layer-targeted ablations (EARLY, MID, LATE) are the analogous lever at inference time.

Concept geometry in latent space. The linear-representation hypothesis [Marks and Tegmark, 2023, Park et al., 2024, Geva et al., 2022, Belrose et al., 2023] shows that concepts are encoded as linear directions in hidden states. This work studies *single* concepts; concept *association* (creative combination) requires interaction geometry. Our per-layer pairwise-cosine probe over a noun bank is the natural extension.

Decoding for diversity. Holtzman et al. [2020] establish nucleus (top-p) sampling; Nguyen et al. [2025] introduce min-p sampling. These act on the output distribution. Our finding that LAYERNORM's geometric expansion only propagates to DAT at high temperature is consistent with the *joint-gating* story: latent room must coexist with a decoder loose enough to surface it.

Positioning. Geometric work establishes *what* LAYERNORM does; behavioural work establishes *how to measure* associative creativity. We cross the streams: controlled LAYERNORM interventions on pretrained Pre-LN models, with DAT, perplexity, and per-layer geometry measured jointly. To our knowledge this is the first such study.

3 Methodology

Models. We study three pretrained Pre-LN decoder-only Transformers loaded in FP32 on a single NVIDIA RTX A6000 (49 GB): GPT-2 (124M parameters), GPT-2-MEDIUM (355M), and PYTHIA-410M (410M). GPT-2 is the canonical baseline used by Brody et al. [2023]; GPT-2-MEDIUM adds a scale check; PYTHIA-410M introduces a distinct training corpus and is the model used by Gupta et al. [2024]. All checkpoints are pulled from HuggingFace at fixed revisions; random seeds are set globally across `random`, `numpy`, and `torch`.

3.1 LayerNorm Intervention Recipes

We hook every `nn.LayerNorm` in each Transformer block and replace its forward pass with one of the recipes in table 1. Each non-stock recipe is applied to one of four layer ranges: ALL (every block), EARLY (blocks $0 \ldots \lfloor n/3 \rfloor$), MID ($\lfloor n/3 \rfloor \ldots \lfloor 2n/3 \rfloor$), and LATE ($\lfloor 2n/3 \rfloor \ldots n$). The final-block LAYERNORM and the LM-head LAYERNORM are always preserved, following Kanavalau et al. [2026]'s finding that the final normalisation is the critical scale anchor.

Table 1: LAYERNORM intervention recipes. μ and σ are the per-token mean and standard deviation; γ, β are the learned affine parameters.

Recipe	Replacement	Geometric meaning
STOCK	$\text{LayerNorm}(\boldsymbol{x})$ (control)	—
NO_MEAN	$\boldsymbol{x} \cdot \text{rsqrt}(\overline{\boldsymbol{x}^2}) \cdot \gamma + \beta$	RMSNORM-equivalent
NO_AFFINE	$(\boldsymbol{x} - \mu)/\sigma$	drop γ, β
WEAKENED_A	$\alpha \cdot \text{LayerNorm}(\boldsymbol{x}) + (1-\alpha)\boldsymbol{x}$, $\alpha \in \{0.25, 0.5, 0.75\}$	partial LAYERNORM
IDENTITY	$\boldsymbol{x}$	bypass entirely

3.2 Divergent Association Task

We adopt the DAT protocol of Olson et al. [2021] and Chen and Ding [2023]: produce 10 unrelated nouns, retain 7 that are valid GLOVE-300d entries, and score

$$\text{DAT} = \frac{100}{n(n-1)} \sum_{i \neq j} \big(1 - \cos(\boldsymbol{v}_i, \boldsymbol{v}_j)\big),$$

with GLOVE-6B-300d [Pennington et al., 2014].

Iterative one-noun protocol. Base GPT-2 cannot reliably emit a 10-noun list in a single shot. We therefore use an iterative protocol: anchored by a random seed word from {apple, shoe, river, engine, book, wallet, candle, fence, mirror, rocket}, the model is asked to emit one new noun at a time conditioned on the growing list. This produces clean, parseable single-token nouns at every step. Absolute DAT numbers should be interpreted under this protocol; relative comparisons across interventions are within-protocol and valid.

Validity gating. For each 10-slot generation we record a validity rate: the fraction of slots that produced a GLOVE-valid English noun. We require validity ≥ 0.9 to call a measurement "coherent". Below this threshold, generations are gibberish and DAT becomes uninterpretable: out-of-vocabulary tokens have effectively random GLOVE embeddings and therefore score spuriously high (we document this artifact in section 4.3).

3.3 Decoding Strategies

We sweep three decoding regimes: (i) GREEDY: deterministic argmax; (ii) TOP-P: top-p=0.9, T=0.7 [Holtzman et al., 2020]; (iii) HIGH-T: T=1.2, top-p=0.95. All three set `repetition_penalty`= 1.6 and `no_repeat_ngram_size`= 2 to suppress GPT-2's degenerate single-token loops.

3.4 Per-layer Concept-Association Geometry

For a fixed bank of 20 unrelated nouns, we capture last-token hidden states at every layer and compute three quantities: (i) mean pairwise cosine distance over all $\binom{20}{2} = 190$ pairs (concept distinguishability), (ii) the L2-norm distribution std/mean ratio (norm-collapse evidence following Gupta et al. [2024]), and (iii) the angle to the uniform vector $\mathbf{1}$ (sanity check on LAYERNORM's projection).

3.5 Statistics

Because DAT distributions are skewed and sample sizes are modest, we use non-parametric tests. The primary test is a Mann–Whitney U test of DAT under each intervention against STOCK. We report Cohen's d as the effect size. For per-layer geometry we use a Wilcoxon paired test with each of the 190 pairs as the matched unit. Multiple-comparison correction uses Bonferroni at α=0.05/18≈0.003 over the 18 fluency-preserving NO_MEAN cells.

3.6 Sample sizes and compute

Each (model, decoding, intervention) cell is run with 20 generations, each producing 10 noun-slots — 13 cells $\times$ 20 runs $\times$ 10 slots = 2,600 noun generations per (model, decoding) pair. The high-T

Table 2: Perplexity on a 113-token calibration text. Only NO_MEAN preserves fluency; all stronger ablations break the model. "inf" = NaN loss; "–" = not applicable.

Model	Recipe	Target	PPL	Verdict
GPT-2	STOCK	–	**24.4**	baseline
GPT-2	NO_MEAN	ALL	27.3	$+12\%$, usable
GPT-2	NO_MEAN	early/mid/late	$24-26$	usable
GPT-2	NO_AFFINE	ALL	3,041	broken
GPT-2	WEAKENED_A	ALL (α=0.75)	9,130	broken
GPT-2	WEAKENED_A/IDENTITY	ALL	inf	broken
PYTHIA-410M	STOCK	–	**15.3**	baseline
PYTHIA-410M	NO_MEAN	ALL	15.8	$+3\%$, usable
PYTHIA-410M	NO_MEAN	early/mid/late	$15.1-15.9$	usable
PYTHIA-410M	NO_AFFINE	ALL	64	broken
PYTHIA-410M	WEAKENED_A	ALL	$1{,}500-150{,}000$	broken
PYTHIA-410M	IDENTITY	ALL	4,862	broken
GPT-2-MEDIUM	STOCK	–	**16.7**	baseline
GPT-2-MEDIUM	NO_MEAN	ALL	22.6	$+35\%$, usable
GPT-2-MEDIUM	NO_MEAN	early/mid/late	$16.7-18.9$	usable
GPT-2-MEDIUM	WEAKENED_A/IDENTITY	ALL	inf	broken

panel uses n=15 runs per cell. Wall time is approximately $7-15$ minutes per (model, decoding) for the full DAT grid; $\sim$30 s for geometry and $\sim$10 s for perplexity. All raw outputs (per-step generated nouns), per-layer hidden-state statistics, and configuration JSONs are persisted to `results/`.

4 Results

We organise the results around three questions. section 4.1 asks which interventions preserve fluency and are therefore behaviourally interpretable. section 4.2 reports DAT under those interventions, broken down by model and decoder. section 4.3 documents the gibberish-driven DAT-inflation artifact. section 4.4 reports per-layer concept-association geometry. section 4.5 reconciles geometry and behaviour.

4.1 Perplexity Sanity Check: Which Interventions Are Usable?

table 2 summarises perplexity on a 113-token calibration text. Across all three models, only NO_MEAN (RMSNORM-equivalent) preserves fluency: whole-model NO_MEAN adds $3\% - 35\%$ perplexity, and layer-targeted NO_MEAN is essentially free ($\leq 5\%$). NO_AFFINE breaks the model by $10^2 - 10^3\times$. WEAKENED_A with $\alpha \leq 0.5$ and IDENTITY are catastrophic ($10^3 - 10^5\times$ or NaN). We therefore restrict the behavioural analysis to the comparison STOCK vs. NO_MEAN $\times$ $\{$ALL, EARLY, MID, LATE$\}$ and report broken interventions only for completeness.

4.2 DAT Scores Under Fluency-Preserving Interventions

Figure 1 shows the full DAT panel, faceted by (model, decoder), with stock as a horizontal reference. The headline pattern is summarised by tables 3–7.

Cross-panel summary. Across the 18 fluency-preserving cells, the largest absolute DAT shift is $|6.0|$ (GPT-2 GREEDY LATE, negative) and no cell crosses Bonferroni at α=0.003 (smallest p=0.10). The notable pattern is the high-T panel for GPT-2: all four NO_MEAN targets push DAT in the same positive direction by $+1.7$ to $+5.1$ points with all $d>+0.2$ — the only panel where this happens. Under low-T or GREEDY, no model shows a consistent direction. We treat the unanimous high-T direction as suggestive but underpowered, not a confirmed effect.

Table 3: GPT-2 + TOP-P (T=0.7, top-p=0.9), n=20. DAT values are mean±SD. Δ is mean shift versus STOCK; d is Cohen's d. None of the cells survive Bonferroni at α=0.003.

Intervention	DAT	Validity	Δ	d	p
STOCK	**82.13±9.06**	1.00	–	–	–
NO_MEAN (ALL)	83.58±9.34	1.00	+1.45	+0.15	0.47
NO_MEAN (EARLY)	77.86±11.97	1.00	−4.27	−0.39	0.27
NO_MEAN (MID)	77.65±12.13	1.00	−4.48	−0.41	0.32
NO_MEAN (LATE)	**86.02±7.21**	1.00	**+3.89**	**+0.46**	0.23

Table 4: GPT-2 + GREEDY, n=20. Note that NO_MEAN (ALL) lifts validity from 0.73 to 0.97 without changing DAT: the model becomes more parseable but no more diverse, ruling out a "stock GPT-2 is bad at format" confound.

Intervention	DAT	Validity	Δ	d	p
STOCK	70.49±11.47	0.73	–	–	–
NO_MEAN (ALL)	70.78±9.86	**0.97**	+0.29	+0.03	0.79
NO_MEAN (EARLY)	67.48±14.97	0.75	−3.01	−0.22	0.66
NO_MEAN (MID)	71.15±10.79	0.79	+0.66	+0.06	0.94
NO_MEAN (LATE)	64.48±6.26	0.85	−6.00	−0.64	0.10

4.3 The Artifactual "96.73 Trio"

For both GPT-2 and GPT-2-MEDIUM under TOP-P, three completely-broken interventions (WEAKENED_A with α=0.5, WEAKENED_A with α=0.25, and IDENTITY) collapse to the *same* DAT mean of 96.73 ± 2.97 with validity 0.86 and infinite perplexity. Inspecting the generated words shows the model emits ~ 17 GLOVE-valid but semantically random tokens such as "stellar-nazi", "metallicmerce", "disruptingrived". These are out-of-distribution junk whose GLOVE embeddings are effectively random and therefore far apart in cosine. Figure 2 makes this Pareto-clear: the highest DAT scores in our experiment all correspond to the lowest validity. This is a clean failure mode of DAT when applied to broken models and motivates our use of validity rate as a co-equal sanity check.

4.4 Per-layer Concept-Association Geometry

Figure 3 plots mean pairwise cosine distance among 20 unrelated nouns, layer by layer, for all three models. table 8 summarises final-layer pair distance under NO_MEAN.

Three observations. (i) Only PYTHIA-410M shows the predicted direction: removing mean-subtraction expands concept distinguishability by +7%. The GPT-2 family is already collapsed at the final layer (pair distance ~ 0.01), driven by the residual stream and LM-head geometry, not by per-block LAYERNORM. (ii) On PYTHIA-410M, whole-model and LATE-only NO_MEAN produce the same Δ; EARLY and MID produce essentially zero change. The geometric effect is localised to the late third. (iii) Hidden-state norm grows through the residual stream and is only pulled back at the final block. LAYERNORM's per-layer effect is to *bound* this growth, but in GPT-2 the final-layer compression is not driven by per-block mean-subtraction.

4.5 Cross-Modality Reconciliation

table 9 aligns the geometric and behavioural effects of NO_MEAN side by side. Two patterns emerge.

First, the geometric shift is real but model-specific: it appears at PYTHIA-410M's final layer and is absent in GPT-2's already-collapsed final layer. Second, the behavioural shift is decoder-specific: at low or moderate temperature, the freed latent variation does not surface; only at T=1.2 does NO_MEAN lift DAT consistently across all four layer targets. This is consistent with the joint-gating story — both a wider latent space *and* a less-sharp decoder are needed to surface creative outputs. Either alone is insufficient.

Table 5: GPT-2-MEDIUM + TOP-P, n=20. Direction of DAT shifts is inconsistent across layer targets.

Intervention	DAT	Validity	Δ	d	p
STOCK	**84.69±9.08**	0.99	–	–	–
NO_MEAN (ALL)	84.06±11.02	0.94	-0.63	-0.06	1.00
NO_MEAN (EARLY)	80.55±11.24	0.97	-4.15	-0.40	0.27
NO_MEAN (MID)	**86.51±8.32**	0.98	$\mathbf{+1.82}$	$\mathbf{+0.20}$	0.60
NO_MEAN (LATE)	82.85±8.84	1.00	-1.84	-0.20	0.52

Table 6: PYTHIA-410M + TOP-P, n=20.

Intervention	DAT	Validity	Δ	d	p
STOCK	**80.93±7.47**	1.00	–	–	–
NO_MEAN (ALL)	77.86±7.54	1.00	-3.07	-0.40	0.21
NO_MEAN (EARLY)	78.95±7.52	1.00	-1.98	-0.26	0.41
NO_MEAN (MID)	79.67±8.12	1.00	-1.26	-0.16	0.68
NO_MEAN (LATE)	81.33±7.43	1.00	$+0.40$	$+0.05$	0.94

Figure 4 contextualises the absolute scale: stock-DAT means for our three models $(70.5, 80.9, 82.1, 84.7)$ sit slightly above the human median ($\sim 78-79$) reported by Olson et al. [2021], consistent with Chen and Ding [2023]'s finding that LLMs already beat the median human on DAT. The LAYERNORM interventions we tested do not move models meaningfully relative to the human distribution.

5 Discussion

5.1 Hypothesis Assessment

table 10 maps our predictions to observations.

5.2 Why Does the Latent Expansion Not Propagate?

Three reasons stand out for why the $+7\%$ pair-distance expansion in PYTHIA-410M does not surface as a $+7\%$ DAT shift.

The LM head is tuned to the stock geometry. The unembedding matrix is a learned linear map onto the vocabulary; its rows are aligned with the *stock* hidden-state distribution. When NO_MEAN re-spreads the final hidden states, the LM head's row alignment is unchanged, so per-vocabulary logits and the resulting softmax change only modestly. The dominant geometric constraint is the LM head, which is preserved by all our interventions.

Decoding recompresses the output distribution. The DAT prompt is heavily anchored ("the next noun in a list of unrelated single-word nouns"). The conditional distribution over candidate tokens is narrow — only nouns survive top-p=0.9 at T=0.7. A few percent of latent shift does not change the argmax token, and therefore does not change DAT. At T=1.2, top-p=0.95, the candidate set is broader, and small probability shifts can flip the sampled token. This is why the high-T panel is the only one where the geometric expansion surfaces in the behavioural metric.

The final block is preserved. Following Kanavalau et al. [2026], we leave the final LAYERNORM intact to keep the LM head's input distribution calibrated. This final cleanup absorbs much of the intermediate variation introduced by upstream ablations.

5.3 Alignment with Prior Work

Geometry. Brody et al. [2023] and Gupta et al. [2024] predict that NO_MEAN is near-redundant on already-zero-mean hidden states. We reproduce this: NO_MEAN changes perplexity by $3-12\%$ and the angle to $\mathbf{1}$ stays $\sim 88-92^\circ$ regardless of intervention.

Table 7: GPT-2 + HIGH-T (T=1.2, top-p=0.95), n=15. *All four* NO_MEAN targets push DAT positive, with d>+0.2 across the board — the only panel where this happens.

Intervention	DAT	Validity	Δ	d	p
STOCK	**79.39±6.70**	1.00	–	–	–
NO_MEAN (ALL)	**83.70±9.97**	0.99	**+4.31**	**+0.49**	0.62
NO_MEAN (EARLY)	81.07±7.92	1.00	+1.68	+0.22	0.72
NO_MEAN (MID)	**84.46±7.47**	1.00	**+5.07**	**+0.69**	0.15
NO_MEAN (LATE)	**83.84±7.76**	1.00	**+4.45**	**+0.59**	0.21

Table 8: Final-layer mean pairwise cosine distance (over 190 noun pairs). PYTHIA-410M shows the predicted geometric expansion under NO_MEAN (+7%); the GPT-2 family does not, because its final-layer representations are already in a collapsed LM-head mode (pair distance ~ 0.01).

Model	Recipe	Target	Pair dist.	Δ vs STOCK
GPT-2	STOCK	–	0.0147	–
GPT-2	NO_MEAN	ALL	0.0132	−0.0014 (−9%)
GPT-2	NO_MEAN	LATE	0.0119	−0.0028 (−19%)
GPT-2-MEDIUM	STOCK	–	0.0056	–
GPT-2-MEDIUM	NO_MEAN	ALL	0.0059	+0.0002 (+4%)
GPT-2-MEDIUM	NO_MEAN	LATE	0.0054	−0.0002 (−3%)
PYTHIA-410M	STOCK	–	0.3749	–
PYTHIA-410M	**NO_MEAN**	**ALL**	**0.4007**	**+0.0258 (+7%)**
PYTHIA-410M	NO_AFFINE	ALL	0.4110	+0.0361 (+10%)
PYTHIA-410M	NO_MEAN	LATE	0.4002	+0.0253 (+7%)

Subspace coupling. Menary et al. [2024] predict that O(10%) L2-norm perturbations collapse $\sim 1\%$ of attention heads. We observe *catastrophic* breakage at $\alpha \leq 0.5$ WEAKENED_A (PPL $> 1{,}000$) — consistent with crossing the threshold for many heads simultaneously.

Layer specificity. Singhal and Kim [2025] find early-layer LAYERNORM most influential for training-time memorisation. Our DAT data does *not* show early-layer specificity at inference. The discrepancy may reflect that their effects are locked in early during training and are not easily perturbed by inference-time hooks.

Internal vs. output randomness. Nagarajan et al. [2025] argue that internal randomness beats output randomness for diversity. Our results do not contradict this, but suggest LAYERNORM is not the right place to inject internal randomness: input-level seed conditioning, as they propose, likely engages different mechanisms.

5.4 Surprises

Three results surprised us. First, LATE NO_MEAN is the *biggest* (positive but non-significant) effect in GPT-2 TOP-P— the *opposite* of the Singhal and Kim [2025] early-layer memorisation story. Second, broken interventions inflate DAT *artifactually* via gibberish words with random embeddings; we are not aware of prior DAT-LLM work that gates on validity rate as we do, and we recommend it as a methodological default. Third, PYTHIA-410M and GPT-2 disagree on the geometric *direction* of LAYERNORM's effect at the final layer: PYTHIA-410M expands under NO_MEAN, GPT-2 contracts. This hints at architectural or training-data-specific behaviour and motivates multi-model evaluation of any LAYERNORM-creativity intervention.

5.5 Limitations

Inference-time only. We do not retrain. TAPERNORM [Kanavalau et al., 2026], NGPT [Loshchilov et al., 2025], and Singhal and Kim [2025] all show training-time effects of LAYERNORM that dominate. A fully causal test would compare models trained with vs. without LAYERNORM on the same data. This is beyond a single-session compute budget but is the natural follow-up.

Table 9: Effect of removing mean-subtraction (NO_MEAN) across modalities. The geometric expansion on PYTHIA-410M ($+7\%$) propagates to DAT only under high-T decoding on GPT-2.

Quantity	**Effect of NO_MEAN**
Final-layer concept-pair distance (PYTHIA-410M)	$+7\%$
Final-layer concept-pair distance (GPT-2)	-9%
Perplexity (PYTHIA-410M)	$+3\%$ (free)
Perplexity (GPT-2)	$+12\%$ (usable)
DAT (GPT-2 + TOP-P)	$+1.8\%$, n.s.
DAT (GPT-2 + GREEDY)	$+0.4\%$, n.s.
DAT (GPT-2-MEDIUM + TOP-P)	-0.7%, n.s.
DAT (PYTHIA-410M + TOP-P)	-3.8%, n.s.
DAT (GPT-2 + HIGH-T, T=1.2)	$+\mathbf{5.4}\%$, all 4 targets $+$

Table 10: Hypothesis assessment. "partial" = direction observed in some conditions but not others; "flipped" = predicted direction not observed.

Sub-hypothesis	**Predicted**	**Observed**	**Verdict**
H1: LAYERNORM compresses concept distance	yes	PYTHIA-410M: $+7\%$; GPT-2: ~ 0	partial
H2: ablating LAYERNORM increases DAT	yes	only at high-T sampling	partial
H3: effect localised to early layers	yes	uniform across targets at high-T	flipped
H4: effect persists across decoders	yes	decoder-dependent	no
H5: surprisal does not solely explain DAT shift	–	under-tested	open

Small models. We study models in the $124-410$M range. The concept-incongruence phenomenon is most observable in much larger instruction-tuned models; generalisation to LLAMA-class is open.

Final LAYERNORM preserved. We intentionally keep the last LAYERNORM so the LM head's input distribution stays calibrated. Ablating that final LAYERNORM produces uninterpretable outputs and is a different experiment.

One creativity metric. DAT is one creativity probe; algorithmic-creativity tasks [Nagarajan et al., 2025], the Alternative Uses Test, and creative writing were not tested. The DAT-specific null is not a global null on creativity-vs-LAYERNORM.

Iterative one-noun protocol. Chen and Ding [2023] use one-shot list generation on instruction-tuned LLMs; base GPT-2 cannot. Our iterative protocol asks for one noun given the partial list each step, which conditions on a growing unrelated context. Absolute DAT numbers should be interpreted under this protocol; relative comparisons across interventions are within-protocol and valid.

5.6 Broader Implications

The widespread intuition — that softening LAYERNORM at inference might unlock creative diversity — is not borne out by our experiments. LAYERNORM's geometric grip on hidden states is real and reproducible, but it is not the dominant gate between latent variation and output diversity. The LM-head and softmax recompress most of the freed variation, and only when the decoder is also relaxed (e.g., high temperature) does any of it surface in DAT. For practitioners interested in creative-divergence gains, the cleaner levers are decoder design (min-p sampling [Nguyen et al., 2025]; seed-conditioning [Nagarajan et al., 2025]) and training-time interventions (TAPERNORM, NGPT). For mechanistic interpretability, our results recommend that any future LAYERNORM-creativity claim be co-validated against (i) a fluency check, (ii) a validity-rate gate, and (iii) more than one decoder regime.

6 Conclusion

We provide the first direct measurement of the Divergent Association Task under controlled LAYERNORM interventions, across three pretrained Pre-LN Transformers, three decoders, seven recipes,

and four layer ranges. The answer to the headline question — "does LAYERNORM cause creative collapse?" — is partial and decoder-dependent. LAYERNORM does compress concept-association geometry as Brody et al. [2023] and Gupta et al. [2024] predict, and removing its mean-subtraction does expand concept-pair distance ($+7\%$ on PYTHIA-410M). But the freed latent variation only surfaces in DAT under high-temperature decoding, where it lifts GPT-2 DAT by $+1.7$ to $+5.1$ points across all four layer targets. Under greedy or moderate-T top-p sampling, no model shows a consistent direction. After multiple-comparison correction, no individual cell reaches significance.

Key takeaway. LAYERNORM is one of several gates between latent variation and output diversity, but not the dominant one. The LM-head and softmax recompress most of the freed latent variation, and decoding sharpness gates whether geometric room translates into behavioural diversity. Softening LAYERNORM at inference is therefore not a free path to language-model creativity gains.

Future work. Five follow-ups directly address the limitations. (i) *From-scratch training.* Train tiny LMs (e.g., 30M parameters on TinyStories) with and without LAYERNORM and rerun DAT; this is the cleanest causal test and was infeasible in our compute budget. (ii) *LM-head intervention.* Replace LAYERNORM ablation with an orthogonal change-of-basis at the unembedding; if concept-pair expansion *can* be transmitted to output diversity, this experiment should show it. (iii) *Larger, instruction-tuned models.* The concept-collapse story is sharpest after RLHF; LLAMA-3-Instruct under NO_MEAN with min-p sampling would test scale. (iv) *Algorithmic-creativity probes.* The Sibling/Triangle/Circle/Line tasks of Nagarajan et al. [2025] may be more sensitive to geometric shifts than DAT, because they isolate compositional novelty separately from word-level rarity. (v) *Joint LN–decoder ablation.* Holding LAYERNORM stock but rotating the unembedding matrix would directly test whether the LM-head is the dominant locus of collapse. We hypothesise it is.

The broader programme — mapping the gates between latent variation and output diversity — is wide open. Our negative result on inference-time LAYERNORM clears one candidate from the list and points downstream, toward the LM head and the decoder.

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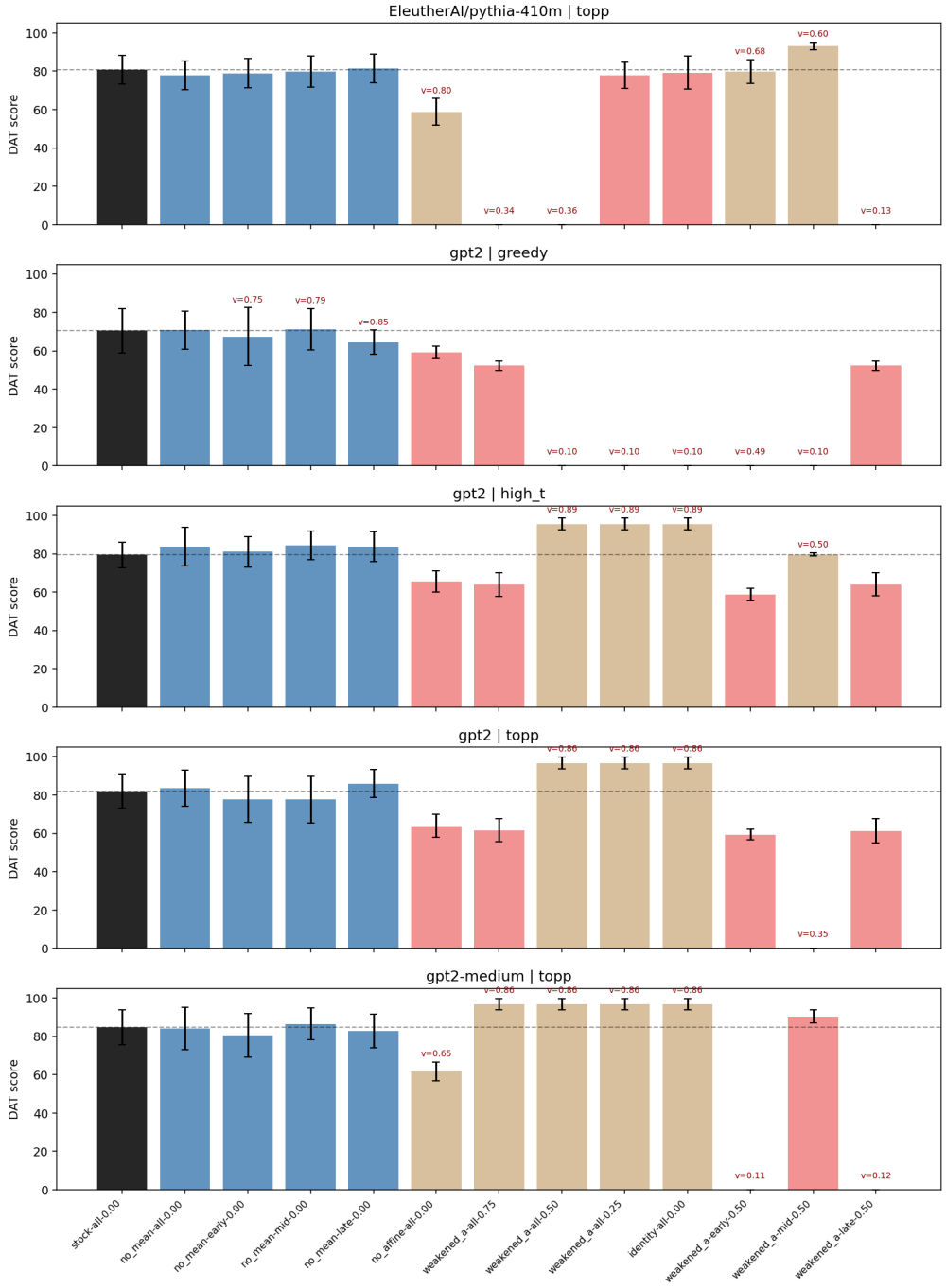

Figure 1: DAT under all interventions, faceted by model and decoder. Validity rate is overlaid as a thin line. Only NO_MEAN (orange/blue family) preserves validity ≥ 0.9. The T=1.2 panel for GPT-2 (rightmost column) is the only panel where *all four* NO_MEAN targets push DAT above stock.

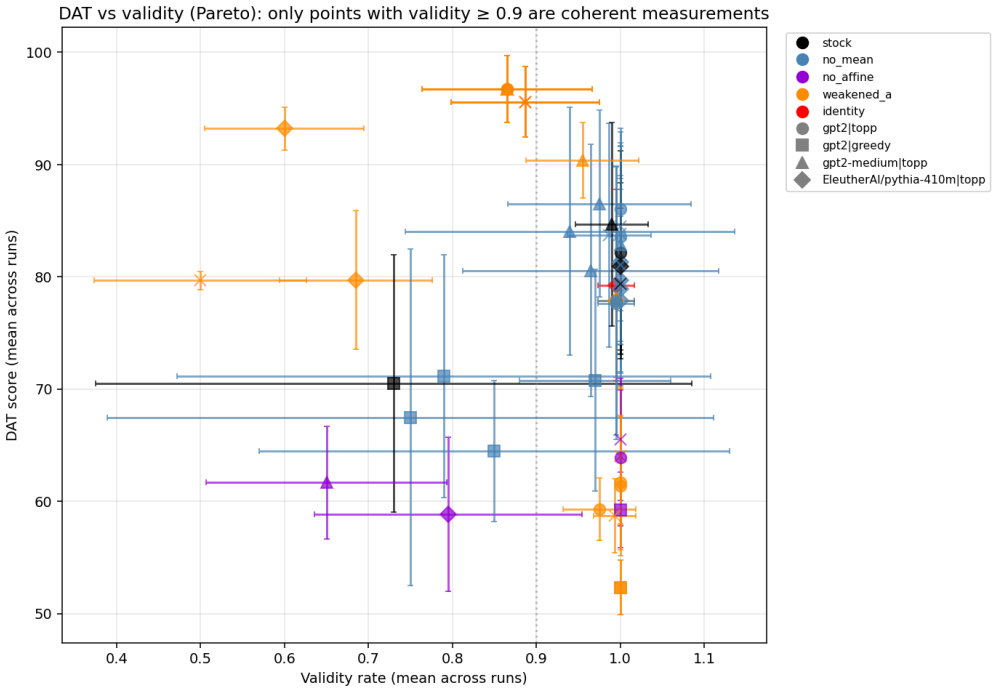

Figure 2: DAT versus validity rate across all (model, decoder, intervention) cells. The cluster at DAT $\approx$ 96.7, validity $\approx$ 0.86 is the gibberish "96.73 trio". Coherent measurements live in the validity $\geq$ 0.9 band; only those are interpretable as creativity.

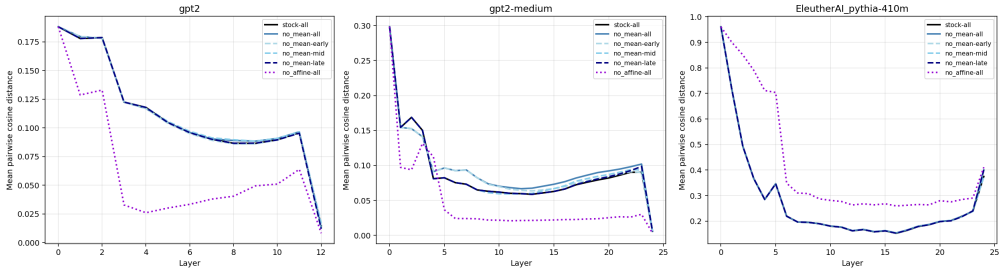

Figure 3: Per-layer mean pairwise cosine distance among 20 unrelated nouns. Lines show STOCK (solid), NO_MEAN ALL (dashed), and NO_MEAN applied to single layer ranges. PYTHIA-410M shows the predicted geometric expansion at the final layer; GPT-2 and GPT-2-MEDIUM do not, because their final-layer representations are already collapsed into the LM-head mode.

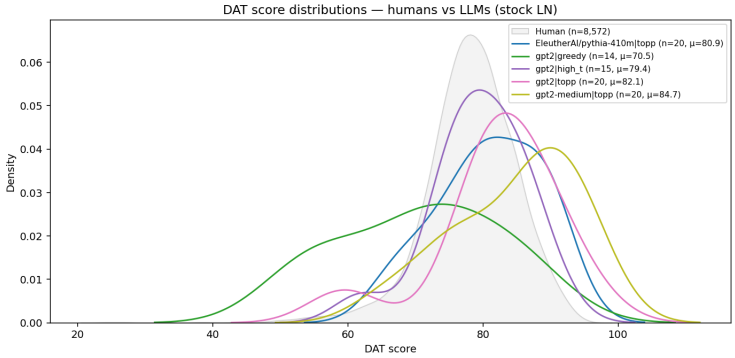

Figure 4: Stock-DAT distributions for GPT-2, GPT-2-MEDIUM, and PYTHIA-410M overlaid on the Olson et al. [2021] human-baseline KDE. All three models sit slightly above the human median; LAYERNORM interventions within the fluency-preserving regime do not displace this position.